

Safety Data Sheets (SDSs)

Product Name: Rechargeable Li-ion Battery

Commissioner: Jiangsu CHO Technology Co., Ltd

CVC Testing Technology Co., Ltd.



Safety Data Sheets (SDSs)

Ref. No.: RZUN012025-7530-DS1

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Name of Product	Rechargeable Li-ion Battery
Type/Model	Battery Li-01 22.2V 2000mAh 44.4Wh
Commissioned by	Jiangsu CHO Technology Co., Ltd
Commissioner address	299 Xinyang Road, Ehu Town, Xishan District, Wuxi City
Supplier	Jiangsu CHO Technology Co., Ltd
Supplier address	299 Xinyang Road, Ehu Town, Xishan District, Wuxi City
Inspection according to	GLOBALLY HARMONIZED SYSTEM OF CLASSIFICATION AND LABELLING OF CHEMICALS (GHS, Rev.11)
Emergency telephone number	18652423886
Remarks	-
 Seal of CVC Date of issue: 2025-12-16	

Approved by: Zhang Siyao

Zhang siyao

Reviewed by: Liu Zhen

liuzhen

Tested by:

Lin Qingyuan

Lin Qingyuan

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SECTION 1: Product and company identification

Product Identifier:

Product name: Rechargeable Li-ion Battery

Model: Battery Li-01

Other means of identification:

Synonyms: None

Relevant identified use of Product and uses advised against:

Recommended Use: -

Uses advised against: -

Details of the Supplier of the safety data sheet:

Name: Jiangsu CHO Technology Co., Ltd

Address: 299 Xinyang Road, Ehu Town, Xishan District, Wuxi City

Telephone: 18652423886

Fax: -

Postcode: -

E-mail address: 361134468@qq.com;hels@choea-china.com

Emergency telephone number:

Company Emergency Phone Number: 18652423886

SECTION 2: Hazard identification

Classification:

The watt-hour rate of the product is 44,4Wh, it is not belong to the lithium-ion battery.

The product is tested according to Section 38.3 of the Manual of Tests and Criteria, the test report number is: RZUN2024-2743

Other information

Caution! Avoid short circuit place in high temperature environment, put into water, or damage the shell.

SECTION 3: Composition/information on ingredients

Chemical characterization: Mixtures

Description: Chemical power supply based on nonaqueous electrolyte. Composed by positive electrode, negative electrode, diaphragm, electrolyte and shell.

Hazardous ingredients:

Common Chemical Name	Chemical Formula	Concentration (%)	CAS No.	EC No.
lithium nickel cobalt manganese oxides	LiCoMnNiO ₄	30.63%	182442-95-1	695-690-9

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graphite	C	18.00%	7782-42-5	231-955-3
lithium hexafluorophosphate	F6LiP	2.25%	21324-40-3	244-334-7
Polyvinylidene fluoride	(C2H2F2)x	0.59%	24937-79-9	607-458-6
Iron	Fe	18.36%	7439-89-6	231-096-4
Copper	Cu	11.11%	7440-50-8	231-159-6
Aluminum	Al	4.47%	7429-90-5	231-072-3
Nickel	Ni	0.83%	7440-02-0	231-111-4
carbon black	C	1.24%	1333-86-4	215-609-9
styrene-butadiene rubber	C12H14	0.67%	9003-55-8	618-370-2
carboxymethyl cellulose sodium salt	C2H4O3.xNa.x Unspecified	0.28%	9004-32-4	618-378-6
electrolyte solvents	-	7.79%	-	-
ethelyne carbonate	C3H4O3		96-49-1	202-510-0
dimethyl carbonate	C3H6O3		616-38-6	210-478-4
propylene carbonate	C4H6O3		108-32-7	203-572-1
ethyl methyl carbonate	C4H8O3		623-53-0	613-014-2
Polyethylene	(C2H4)n	2.25%	9002-88-4	618-339-3
Polyethylene terephthalate	(C10H8O4)n	1.53%	25038-59-9	607-507-1

Note: N/A=Not apply.

SECTION 4: First-aid measures

First aid measures:

Eye Contact: Rinse thoroughly with plenty of water, also under the eyelids. If symptoms persist, call a physician.

Skin Contact: Remove contaminated clothing and shoes. Wash skin with soap and water. In the case of skin irritation or allergic reactions see a physician.

Inhalation: Move to fresh air. If symptoms persist, call a physician.

Ingestion: Do NOT induce vomiting. Drink plenty of water. If symptoms persist, call a physician.

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Swallowing: Do not induce vomiting. Get medical attention.

Most Important Symptoms/Effects:

No information available.

Indication of any immediate medical attention and special treatment needed:

Inform physician. Treat symptomatically.

SECTION 5: Fire-fighting measures

Suitable Extinguishing Media:

CO₂, dry chemical powder, wet sand, plenty of water (for cooling).

Unsuitable Extinguishing Media: No information available.

Protective Equipment and Precautions for Firefighters:

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. For example: Wear self-contained respiratory protective device. Wear suitable protective clothing and eye/face protection.

Special hazards arising from the substance or mixture:

Battery may burst and release hazardous decomposition products when exposed to a fire situation. Lithium batteries contain flammable ingredients that may vent, ignite and produce sparks when subjected to high temperature (>150°C), When damaged or abused (e.g. mechanical damage or electrical overcharging); may burn rapidly with flare-burning effect; may ignite other batteries in close proximity.

SECTION 6: Accidental release measures

Personal precautions, protective equipment and emergency procedures:

Personal Precautions Avoid contact with eyes.

Refer to section 8 for personal protective equipment. Ensure adequate ventilation. Remove all sources of ignition.

Evacuate personnel to safe areas.

Environmental precautions:

Environmental Precautions Refer to protective measures listed in Sections 7 and 8.

Absorb with liquid-binding material (sand, diatomite, acid binders, universal binders, sawdust).

Dispose contaminated material as waste according to item 13.

Methods and material for containment and cleaning up:

Methods for Containment Prevent further leakage or spillage if safe to do so.

Methods for Cleaning up Use personal protective equipment. Dam up. Cover liquid spill with sand, earth or other Non-combustible absorbent material. Pick up and transfer to properly labeled containers. Clean contaminated surface thoroughly.

SECTION 7: Handling and storage

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Precautions for safe handling:

Keep away from ignition sources, heat and flame. Such batteries must be packed in inner packages in such a manner as to effectively prevent short circuits and to prevent movement which could lead to short circuits. Avoid mechanical or electrical abuse.

More than a momentary short circuit will generally reduce the battery service life. Avoid reversing battery polarity within the battery assembly. In case of a battery unintentionally be crushed, rubber gloves must be used to handle all battery components. Avoid contact with eyes, skin. Avoid inhalation. No smoking at working site. Materials to Avoid: Strong oxidizing agents, Corrosives.

Conditions for safe storage, including any incompatibilities:

Store in a cool, well-ventilated area. Keep away from ignition sources, heat and flame. Such batteries must be packed in inner packages in such a manner as to effectively prevent short circuits and to prevent movement which could lead to short-circuits. Materials to Avoid: Strong oxidizing agents, Corrosives.

SECTION 8: Exposure controls/personal protection

Engineering Controls:

Use ventilation equipment if available. Safety shower and eye bath.

Personal Protective Equipment:

Respiratory System: Not necessary under conditions of normal use.

Eyes: Not necessary under conditions of normal use.

Clothing: Wear appropriate protective clothing.

Hand: Safety gloves.

Other Protect:

No smoking, drinking and eating at working site. Wash thoroughly after handling.

SECTION 9: Physical and chemical properties

Physical State	Form: Irregular shape	
	Color: Black	
	Odour: Odourless	
	Odor Threshold: No information available	
Change in condition:		
pH, with indication of the concentration		Not determined.
Melting point/freezing point		Not determined.
Initial boiling point and Boiling range:		Not determined.
Flash Point		Not determined.

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Flammability (solid, gas)	Not determined.
Upper/lower flammability or explosive limits	Not determined.
Auto-ignition temperature	Product is not self-igniting.
Decomposition temperature	Not determined.
Other Information	No further relevant information available.

SECTION 10: Stability and reactivity

Reactivity: Stable under recommended storage and handling conditions (see section 7).

Chemical stability: Stable under normal conditions of use, storage and transport.

Thermal decomposition/conditions to be avoided: No decomposition if used according to specifications.

Possibility of Hazardous Reactions: None under normal processing.

Hazardous Polymerization: Hazardous polymerization does not occur.

Conditions to avoid: Strong heating, fire, Incompatible materials.

Incompatible materials: Strong oxidizing agents. Strong acids.

Hazardous Decomposition Products: Carbon oxides, other irritating and toxic gases.

SECTION 11: Toxicological information

Acute toxicity: No data available.

Skin corrosion/irritation: No irritant effect.

Serious eye damage/irritation: Cause serious eye irritation.

Respiratory or skin sensitization: No sensitizing effects known.

Specific target organ system toxicity: No information available.

Note: The internal battery materials may cause irritation to eyes and skin.

SECTION 12: Ecological information

Toxicity: No further relevant information available.

Persistence and degradability: No further relevant information available.

Bioaccumulative potential: No further relevant information available.

Mobility in soil: No further relevant information available.

Results of PBT and vPvB assessment:

PBT: Not applicable.

vPvB: Not applicable.

Other adverse effects: No information available.

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SECTION 13: Disposal considerations

Waste treatment methods:

Recommendation: Lithium batteries are best disposed of as a non-hazardous waste when fully or mostly discharged. Contact a licensed professional waste disposal service to dispose of large quantities materials.

Other disposal recommendations:

Recommendation: Disposal must be made according to official regulations.

SECTION 14: Transport information

The product had been tested according to the requirements of the UN manual of tests and Criteria, Part III, subsection 38.3 (see section 2)

EmS No: F-A ,S-I

Marine pollutant: No

Environmental hazards: Not applicable.

Special precautions for user: Not applicable.

Proper Shipping name: Lithium ion batteries packed with equipment

Hazard Class: Not applicable

UN/ID Number: UN3481

Packaging Group: Not applicable

The package has passed 1.2m drop test. Test report number is: RZUN2024-2743-DT1.

Air transport:

Label for conveyance: Battery Mark

The goods are complied with the requirements of Section II of Packing Instruction 966 of 67th DGR Manual of IATA (2026 Edition).

Maritime transport:

Label for conveyance: Lithium or sodium ion battery mark

The goods are complied with the requirements of Special provision 188 of IMDG CODE (Amdt. 42-24) (2024 Edition).

SECTION 15: Regulatory information

International Regulation:

GLOBALLY HARMONIZED SYSTEM OF CLASSIFICATION AND LABELLING OF CHEMICALS (GHS)

Recommendations on the TRANSPORT OF DANGEROUS GOODS Model Regulations

LTC-R-6037-SDSs C&E- A0

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IATA Dangerous Goods Regulations (DGR)

INTERNATIONAL MARITIME DANGEROUS GOODS CODE (IMDG CODE)

Regulations concerning the International Carriage of Dangerous Goods by Rail (RID)

EU Regulation:

Regulation (EU) 2020/878: Revised Requirements for EU Safety Data Sheets

EU regulation (EC) 1272/2008 on "Classification, Labeling and Packaging of Substances and Mixtures" (CLP)

Registration, Evaluation and Authorization of Chemicals (REACH)

Agreement concerning the International Carriage of Dangerous Goods by Road (ADR)

US Regulation:

American National Standard for Hazardous Workplace Chemicals – Hazard Evaluation and Safety Data Sheet and Precautionary Labeling Preparation

US DOT, Code of Federal Regulations, Title 49, Transportation, PT. 100-185

SECTION 16: Other information

This file is only effective to the batteries(Battery Li-01) provided by commissioner Jiangsu CHO Technology Co., Ltd. The commissioner provides the composition information of batteries, and promises its integrity and accuracy. Users should read this file carefully, and use the batteries in correct method. CVC Testing Technology Co., Ltd. (CVC) doesn't assume responsibility for any damage or loss because of misuse of batteries.

Important

1. The test report is invalid without the official seal of CVC.
2. Nobody is allowed to photocopy or partly photocopy this test report without written permission of CVC.
3. The test report is invalid without the signatures of Ratifier, Reviewer and Testing engineer.
4. The test report is invalid if altered.
5. Objections to the test report must be submitted to CVC within 15 days.
6. This report is valid for the samples provided by commissioner only.

**The test data and test results given in this test report should only be used for purposes of scientific research, teaching and internal quality control when the CMA symbol is not presented. **

Address: No.3, Tiantai 1st Road, Kaitai Avenue, Science City, Guangzhou, Guangdong, China.

Tel: 020 32293888

FAX: 020 32293889

Post Code: 510663

E-mail: office@cvc.org.cn

<http://www.cvc.org.cn>